

Estimated position sequence

$$\hat{\mathbf{P}} \in \mathbb{R}^{T \times 3}$$

Linear output layer

$$d_{\text{model}} = 128 \rightarrow 3$$

Add & layer normalization

Feed-forward network

$$d_{\text{ff}} = 512, \text{ReLU}$$

Add & layer normalization

Multi-head self-attention

$$N_h = 8, d_k = d_v = 16$$

$\times 9$

encoder dropout
0.0534

Sinusoidal positional encoding

sequence length $T = 20$; dropout = 0.0534

Linear embedding

$$F_{\text{in}} \rightarrow d_{\text{model}} = 128; \text{scaled by } \sqrt{d_{\text{model}}}$$

Input feature sequence

$$\mathbf{X} \in \mathbb{R}^{T \times F_{\text{in}}}, T = 20, F_{\text{in}} = 25 \text{ in simulation}$$

Transformer encoder layer